LoopDelta · Stage 0 check · Ouro-1.4B

Does the cross-loop change Δ predict question difficulty?

LoopDelta uses Δ to set per-question depth and to weight votes. Stage 0 tests whether Δ carries that signal.

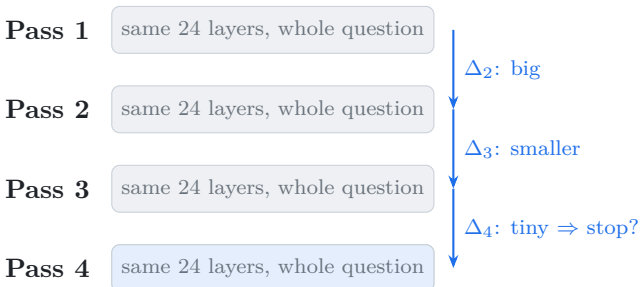
Answer: no

Δ tracks question length, not difficulty. Adaptive depth still has large headroom.

November 2026

The idea

Ouro runs 24 shared layers **4 times**. Δ = change in the KV cache between passes.



Use 1: depth

Stop once Δ is small.

Use 2: voting

Weight solvers by $1/\Delta$.

$\Delta_t = \|X_t - X_{t-1}\|^2 / \|X_{t-1}\|^2$, $X \in \{K, V\}$, averaged over layers.

What we ran

A: prefill only

- 1,000 MATH train questions, 200 per level 1–5
- Also run at 8 loops (2× training)

B: generation

- 300 questions, 60 per level, at 1, 2, 3 and 4 loops
- Per depth: 1 greedy + 4 sampled answers
- 6,000 graded answers

1,000
questions (A)

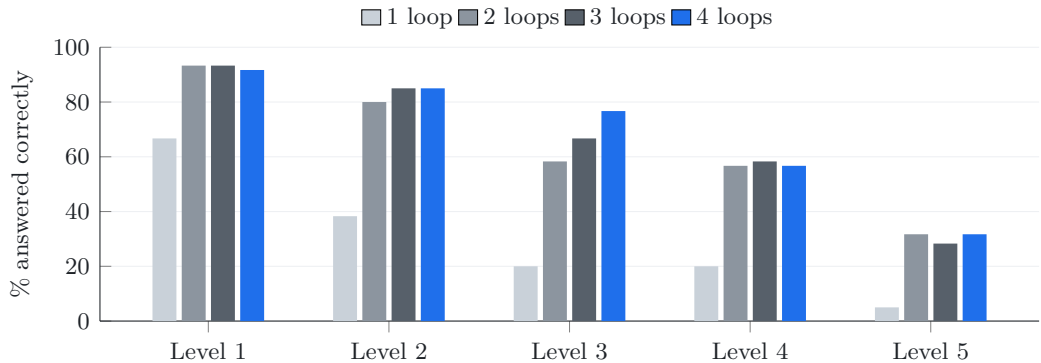
6,000
answers (B)

5
difficulty levels

1–4
loops tested

Fewer loops is often enough

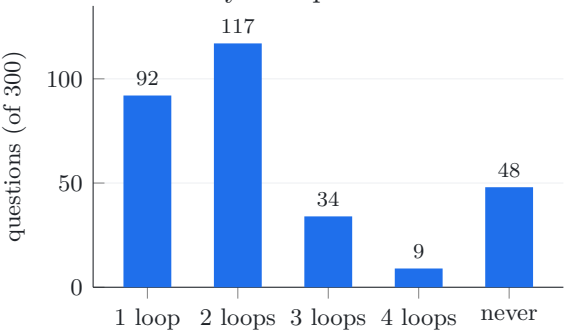
Most of the gain is 1 → 2 loops.



Greedy, 60 questions/level. Overall: 30%, 64%, 66%, 68%. Ouro-1.4B (plain model), MATH train set, run 2026-11-05.

Minimum loops needed per question

70% are solved by 2 loops.



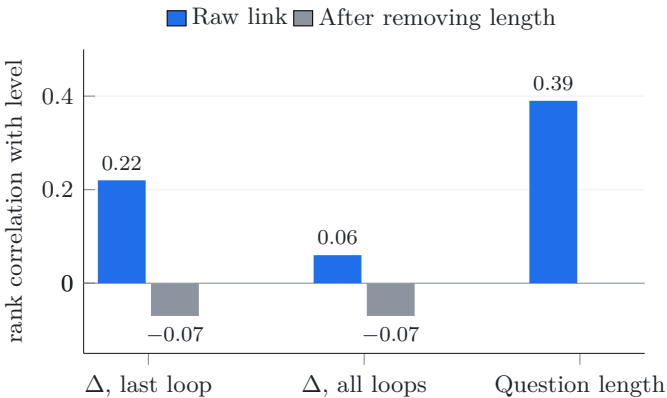
77.8%
oracle depth picker

68.0%
always 4 loops

Fewest loops with a correct greedy answer. Oracle on 153 held-out questions. Ouro-1.4B (plain model), MATH train set, run 2026-11-05.

Test 1: does Δ track difficulty?

Only through question length.



Takeaway

Given length, Δ adds nothing.

8 loops: 0.23 raw, −0.06 partial.

Spearman with level, 1,000 questions, 4 loops. Partial = controlling for length. Ouro-1.4B (plain model), MATH train set, run 2026-11-05.

Test 2: choosing depth

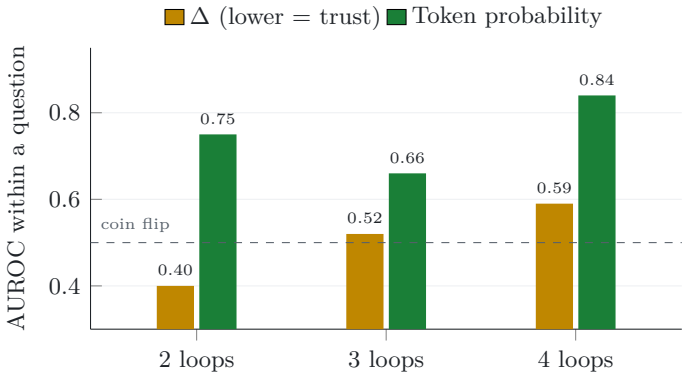
Question length does as well as Δ.

How depth is chosen	Accuracy	Verdict
Always 4 loops	68.0%	baseline
Random	67.5%	chance
Pick by Δ	70.6%	+2.6
Pick by length	70.6%	**same as Δ**
Oracle	77.8%	ceiling

Greedy, thresholds fit on half, scored on 153 held-out questions (1 question = 0.65 pts). Ouro-1.4B (plain model), MATH train set, run 2026-11-05.

Test 3: weighting votes

Δ barely beats chance at ranking right over wrong answers.



Vote accuracy, 4 loops

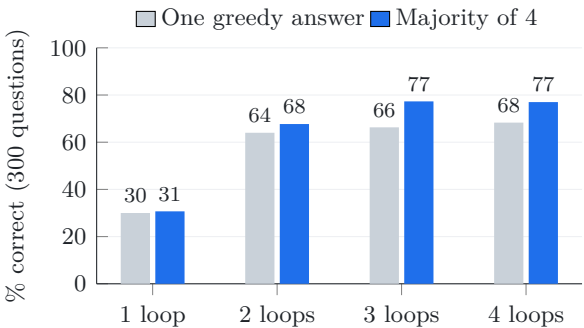
Plain majority	77.0%
Δ-weighted	76.3%
Probability-weighted	76.3%

In 220 of 300 questions all 4 answers are right or all wrong, so weights can't change the vote.

AUROC over right/wrong answer pairs within a question (254–276 pairs from the ∼80 questions with mixed answers). Entropy ≈ token probability. Ouro-1.4B (plain model), MATH train set, run 2026-11-05.

Side finding: voting

Voting: 3 loops = 4 loops



77.3%

vote, 3 loops

77.0%

vote, 4 loops

Voting adds ∼10 points from 3 loops on.

Verdict

Question	Evidence	Answer
Does Δ know difficulty?	$0.22 \rightarrow -0.07$ given length	**only via length**
Does Δ pick depth well?	ties question length	**no**
Does Δ pick good answers?	AUROC 0.40–0.59	**no**
Is adaptive depth worth it?	oracle 77.8% vs 68.0%	**yes**

Next

Drop Δ. Test other depth signals on existing data: length, first-loop entropy, Ouro's exit gate.

Caveats: one model, one dataset; 153 held-out questions; only 3 Δ values per question.